

4 A2 - Temă seminar 7

1. $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$$T(x, y, z) = (x + 2y, 2y, -2y + z)$$

T apl liniară (x matriceală)

$$T(x) = Ax, \text{ unde } x = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \text{ unde } A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ 0 & -2 & 1 \end{pmatrix} \in M_3(\mathbb{R})$$

fie $x_1, x_2 \in \mathbb{R}^3$

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} \quad \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$$

unde $\alpha_1, \alpha_2 \in \mathbb{R}$

$$\begin{aligned} T(\alpha_1 x_1 + \alpha_2 x_2) &= A(\alpha_1 x_1 + \alpha_2 x_2) = A(\alpha_1 x_1) + A(\alpha_2 x_2) = \\ &= (A \alpha_1) x_1 + (A \alpha_2) x_2 = (\alpha_1 A) x_1 + (\alpha_2 A) x_2 = \alpha_1 (A x_1) + \\ &+ \alpha_2 (A x_2) = \alpha_1 T(x_1) + \alpha_2 T(x_2) \Rightarrow T \text{ apl lin} \end{aligned}$$

2. $\mathcal{A}: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$$\mathcal{A}(x, y, z) = (2x - y + 2z, -x + 2y - z, x + y + z)$$

a) $A_{\mathcal{A}} = \begin{pmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{pmatrix} \in M_3(\mathbb{R})$

$\begin{matrix} \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } \end{matrix}$
 $\begin{matrix} x & y & z \\ \mathcal{A}(e_1) & \mathcal{A}(e_2) & \mathcal{A}(e_3) \end{matrix}$

b) $P(\lambda) = \det(A_{\mathcal{A}} - \lambda I_3) = \begin{vmatrix} 2-\lambda & -1 & 2 \\ -1 & 2-\lambda & -1 \\ 1 & 1 & 1-\lambda \end{vmatrix} =$

$$= -\lambda(\lambda-2)(\lambda-3)$$

$$p(\lambda) = 0 \Leftrightarrow -\lambda(\lambda-2)(\lambda-3) = 0 \Rightarrow$$

$$\Rightarrow \begin{cases} \lambda_1 = 0 \\ \lambda_2 = 2 \in \mathbb{R} \\ \lambda_3 = 3 \end{cases} \quad \text{Spec}(\lambda) = \{0, 2, 3\}$$

$$m_0(\lambda_1) = m_2(\lambda_2) = m_3(\lambda_3) = 1$$

Subspace problem:

$$S_{\lambda_1=0} \begin{cases} 2x - y + 2z = 0 \\ -x + 2y - z = 0 \\ x + y + z = 0 \end{cases} \Rightarrow \text{is lin omogen} \\ \text{3 ec, 3 rec}$$

$$\text{rang}(A\lambda - \lambda_1 I_3) = 0$$

$$A\lambda = \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 3 \neq 0 \Rightarrow \begin{cases} x, y \text{ rec wrt} \\ z = \alpha \text{ rec rec} \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} 2x - y = -2\alpha \\ -x + 2y = \alpha \end{cases} \Rightarrow \begin{cases} x = -\alpha \\ y = 0 \end{cases}$$

$$\text{deci } V_{\lambda_1} = \{(-\alpha, 0, \alpha) \mid \alpha \in \mathbb{R}\} = \{ \alpha \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \mid \alpha \in \mathbb{R} \} \\ \stackrel{n}{V_1}$$

$$V_{\lambda_1} = \mathbb{R} \cdot v_1$$

$$S_{\lambda_2=2} \begin{cases} 2x - y + 2z = 2x \\ -x + 2y - z = 2y \\ x + y + z = 2z \end{cases} \Rightarrow \begin{cases} -y + 2z = 0 \\ -x - z = 0 \\ x - y + z = 0 \end{cases}$$

$$A\lambda = \begin{vmatrix} 0 & -1 \\ -1 & 0 \end{vmatrix} = 1 \neq 0 \Rightarrow \text{rang}(A\lambda - \lambda_2 I_3) = 2$$

$$\Rightarrow \begin{cases} x, y \text{ free} \\ z = \beta \text{ free free} \end{cases}$$

$$\Rightarrow \begin{cases} z = 2\beta \\ x = -\beta \end{cases} \Rightarrow$$

$$\text{defi } v_{x2} = \underbrace{\beta(-1, 2, 1)}_{v_2} \mid \beta \in \mathbb{R}$$

$$v_{x2} = \text{span}(v_2)$$

$$S_{\lambda_3=3} \begin{cases} -x - y + 2z = 0 \\ -x - y - z = 0 \\ x + y - 2z = 0 \end{cases}$$

$$\Delta_p = \begin{vmatrix} -1 & 2 \\ -1 & -1 \end{vmatrix} = 3 \neq 0 \Rightarrow \text{rg}(A - \lambda_3 I_3) = 2$$

$$\Rightarrow \begin{cases} x, z \text{ free} \\ y = \tau \text{ free free} \end{cases}$$

$$\begin{cases} -x + 2z = \tau \\ -x - z = \tau \end{cases} \Rightarrow \begin{cases} z = 0 \\ x = -\tau \end{cases} \Rightarrow$$

$$\text{defi } v_{x3} = \underbrace{\tau(-1, 1, 0)}_{v_3} \mid \tau \in \mathbb{R}$$

$$v_{x3} = \text{span}(v_3)$$

c) stud de este diag:

$$m_a(x_1) = m_a(x_2) = m_a(x_3) = 1$$

$$\left. \begin{aligned} v_{\lambda_1} &= \eta \langle v_1 \rangle \Rightarrow \dim v_{\lambda_1} = 1 = m_a(x_1) \\ v_{\lambda_2} &= \eta \langle v_2 \rangle \Rightarrow \dim v_{\lambda_2} = 1 = m_a(x_2) \\ v_{\lambda_3} &= \eta \langle v_3 \rangle \Rightarrow \dim v_{\lambda_3} = 1 = m_a(x_3) \end{aligned} \right\} \quad \Rightarrow$$

matr a este diag

$$d) \quad D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$B = \{ v_1 = (-1, 0, 1), v_2 = (-1, 2, 1), v_3 = (-1, 1, 0) \}$$

1. A) $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$$f(x, y, z) = (-x + 3y - z, -3x + 5y - z, -3x + 3y + z)$$

a) $A_f = \begin{pmatrix} -1 & 3 & -1 \\ -3 & 5 & -1 \\ -3 & 3 & 1 \end{pmatrix} \in M_3(\mathbb{R})$

$\begin{matrix} \text{"} & \text{"} & \text{"} \\ f(e_1) & f(e_2) & f(e_3) \end{matrix}$

b) $p(\lambda) = \det(A_f - \lambda I_3) = \begin{vmatrix} -1-\lambda & 3 & -1 \\ -3 & 5-\lambda & -1 \\ -3 & 3 & 1-\lambda \end{vmatrix} =$

$$= -\lambda^3 + 5\lambda^2 - 8\lambda + 4 = -\lambda^3 + 2\lambda^2 + 3\lambda^2 - 6\lambda - 2\lambda + 4 =$$

$$= -\lambda^2(\lambda - 2) + 3\lambda(\lambda - 2) - 2(\lambda - 2) =$$

$$= (\lambda - 2)(-\lambda^2 + 3\lambda - 2) = -(\lambda - 2)(\lambda - 2)^2$$

$$p(\lambda) = 0 \Leftrightarrow -(\lambda - 2)(\lambda - 2)^2 = 0 \Rightarrow$$

$\Rightarrow \begin{cases} \lambda_1 = 1 \\ \lambda_2 = 2 \end{cases} \in \mathbb{R} \quad \text{Spec}(f) = \{1, 2\}$

$$m_a(\lambda_1) = 1$$

$$m_a(\lambda_2) = 2$$

Luizati proprii:

$$S_{\lambda_1=1} \begin{cases} -2x + 3y - z = 0 \\ -3x + 4y - z = 0 \\ -3x + 3y = 0 \end{cases}$$

$$\Delta_A = \begin{vmatrix} -2 & 3 \\ -3 & 4 \end{vmatrix} = 1 \neq 0 \Rightarrow \text{rg}(A - \lambda_1 I_3) = 2 \Rightarrow$$

$$\Rightarrow \begin{cases} x, y \text{ free} \\ z = x \end{cases}$$

$$\begin{cases} 3y = x + 2x \\ 3y = 3x \end{cases} \Rightarrow \begin{cases} x = x \\ y = x \end{cases}$$

$$\text{donc } v_{\lambda_1} = \{ x(1, 1, 1) / x \in \mathbb{K} \} \Rightarrow v_{\lambda_1} = \text{span}\{v_1\}$$

$$S_{\lambda_2} = 2 \begin{cases} -3x + 3y - z = 0 \\ -3x + 3y - z = 0 \\ -3x + 3y - z = 0 \end{cases} \Rightarrow \begin{cases} \text{rg}(A - \lambda_2 I_3) = 1 \\ -3x + 3y - z = 0 \\ z = -3x + 3y \end{cases}$$

$$\begin{cases} z \text{ free} \\ x = p \\ y = r \end{cases} \Rightarrow z = -3p + 3r$$

$$\text{donc } v_{\lambda_2} = \{ p(1, 0, -3) + r(0, 1, 3), p, r \in \mathbb{K} \}$$

$$v_{\lambda_2} = \text{span}\{v_2, v_3\}$$

c) stud de este diag

$$v_{\lambda_1} = \text{span}\{v_1\} \Rightarrow \dim v_{\lambda_1} = 1 = m_a(\lambda_1)$$

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -3 & 3 \end{pmatrix} \Rightarrow \Delta_A = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \Rightarrow \text{rg} A = 2 \Rightarrow$$

$$v_{\lambda_2} = \text{span}\{v_2, v_3\} \Rightarrow \dim v_{\lambda_2} = 2 = m_a(\lambda_2)$$

$$d) D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, P = \{ v_1 = (1, 1, 1), v_2 = (1, 0, -3), v_3 = (0, 1, 3) \}$$

$$i) \quad f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$f(x, y, z) = (6x - 5y - 3z, 3x - 2y - 2z, 2x - 2z)$$

$$a) \quad A_f = \begin{pmatrix} 6 & -5 & -3 \\ 3 & -2 & -2 \\ 2 & 0 & -2 \end{pmatrix} \in M_3(\mathbb{R})$$

$$b) \quad p(x) = \det(A_f - x I_3) = \begin{vmatrix} 6-x & -5 & -3 \\ 3 & -2-x & -2 \\ 2 & 0 & -2-x \end{vmatrix} =$$

$$= -x^3 + 4x^2 - 5x + 2 = -x^3 + x^2 + 3x^2 - 3x + 2x + 2 =$$

$$= -x^2(x-1) + 3x(x-1) - 2(x-1) = -(x-1)^2(x-2)$$

$$p(x) = 0 \Rightarrow \begin{cases} x_1 = 1 & ; \text{ m.a. } (x_1) = 2 \\ x_2 = 2 & ; \text{ m.a. } (x_2) = 1 \end{cases} \quad \lambda_{\text{pec}} = \{1, 2\}$$

Subspații proprii:

$$\lambda_1 = 1 \quad \begin{cases} 5x - 5y - 3z = 0 \\ 3x - 3y - 2z = 0 \\ 2x - 2y - z = 0 \end{cases}$$

$$\Delta_x = \begin{vmatrix} -3 & -2 \\ -2 & -1 \end{vmatrix} = 1 \neq 0 \Rightarrow \text{rg}(A_f - \lambda_1 I_3) = 2$$

$$\begin{cases} y, z \text{ nec. prime} \\ x = \alpha \text{ nec. nec.} \end{cases}$$

$$\begin{cases} 3y + 2z = 3\alpha & \Rightarrow y = \alpha & \Rightarrow \\ 2y + z = 2\alpha & z = 0 \end{cases}$$

$$\text{deci } V_{\lambda_1} = \{ \alpha (1, 1, 0) / \alpha \in \mathbb{R} \}$$

$$S_{\lambda_2=2} \begin{cases} 4x - 5y - 3z = 0 \\ 3x - 4y - 2z = 0 \\ 2x - 2y - 2z = 0 \end{cases}$$

$$A_{\lambda_2} = \begin{pmatrix} 4 & -5 \\ 3 & -4 \end{pmatrix} = -1 \neq 0 \rightarrow \text{rg}(A_{\lambda_2} - \lambda_2 I_2) = 2$$

$$\begin{cases} x, y \text{ free} \\ z = 0 \end{cases}$$

$$\begin{cases} 4x - 5y = 3z \\ 3x - 4y = 2z \end{cases} \Rightarrow \begin{matrix} -y = -7x \Rightarrow y = 7x \\ x = 8x \end{matrix} \quad \begin{matrix} z = 13 \\ x = 213 \end{matrix}$$

deci

$$V_{\lambda_2} = \{ p(2, 1, 1) \mid p \in \mathbb{R} \}$$

e) stud diag

$$V_{\lambda_1} = \text{span}\{v_1\} \Rightarrow \dim V_{\lambda_1} = 1 \neq m_A(\lambda_1) = 2$$

\(\Rightarrow\) matrice \(A\) nu este diag

71 $\mathcal{A}: \mathbb{R}^4 \rightarrow \mathbb{R}^4$

$$\mathcal{A}(x, y, z, t) = (x + y + z + t, x + y - z - t, x - y + z - t, x - y - z + t)$$

$$A_{\mathcal{A}} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \in M_4(\mathbb{R})$$

$$\mathcal{A}(e_1) \quad \mathcal{A}(e_2) \quad \mathcal{A}(e_3) \quad \mathcal{A}(e_4)$$

$$b) \quad p(\lambda) = \det(A - \lambda I_4) = \begin{vmatrix} 1-\lambda & 1 & 1 & 1 \\ 1 & 1-\lambda & -1 & -1 \\ 1 & -1 & 1-\lambda & -1 \\ 1 & -1 & -1 & 1-\lambda \end{vmatrix} =$$

$$= \lambda^4 - 4\lambda^3 + 16\lambda^2 - 16 = \lambda^4 + 2\lambda^3 - 6\lambda^3 + 12\lambda^2 + 12\lambda^2 + 24\lambda - 8\lambda - 16 =$$

$$= \lambda^3(\lambda+2) - 6\lambda^2(\lambda+2) + 12\lambda(\lambda+2) - 8(\lambda+2) =$$

$$= (\lambda+2)(\lambda^3 - 6\lambda^2 + 12\lambda - 8) = (\lambda+2)(\lambda-2)^3$$

$$p(\lambda) = 0 \Leftrightarrow \begin{cases} \lambda_1 = -2, & m_a(\lambda_1) = 1 \\ \lambda_2 = 2, & m_a(\lambda_2) = 3 \end{cases} \quad \text{Spec} = \{-2, 2\}$$

$$S_{\lambda_1 = -2} = \begin{cases} 3x + y + z + t = 0 \\ x + 3y - z - t = 0 \\ x - y + 3z - t = 0 \\ x - y - z + 3t = 0 \end{cases}$$

$$A_{\lambda_1} = \begin{vmatrix} 3 & 1 & 1 \\ 1 & 3 & -1 \\ 1 & -1 & 3 \end{vmatrix} = 16 \neq 0 \Rightarrow \text{rang}(A - \lambda_1 I_4) = 3$$

$$\begin{cases} x, y, z & \text{rec} \text{ for} \\ t = \alpha & \text{rec} \text{ rec} \end{cases}$$

$$\begin{cases} 3x + y + z = -\alpha \\ x + 3y - z = \alpha \\ x - y + 3z = \alpha \end{cases} \Rightarrow \begin{cases} 4x + 4y = 0 \\ 4x = -4y \\ x = -y \\ y = z \end{cases} \Rightarrow \begin{cases} x = -\alpha \\ y = \alpha \\ z = \alpha \end{cases}$$

$$\text{dec} \quad v_{\lambda_1} = \{ \alpha(-1, 1, 1, 1) / \alpha \in \mathbb{R} \}$$

$$\text{S } \lambda_2 = 2 \quad \begin{cases} -x + y + z + t = 0 & 1-1 \\ x - y - z - t = 0 \\ x - y - z - t = 0 \\ x - y - z - t = 0 \end{cases} \Rightarrow \begin{cases} x - y - z - t = 0 \\ \text{rg}(A - \lambda_2 I_4) = 1 \end{cases}$$

$$\begin{cases} x \text{ rec } \text{rec} \\ y = m, z = r, t = \theta \text{ rec rec} \end{cases} \Rightarrow x = m + r + \theta$$

$$\Rightarrow \text{dec } v_{\lambda_2} = \{ m \underset{v_2}{(1, 1, 0, 0)} + r \underset{v_3}{(1, 0, 1, 0)} + \theta \underset{v_4}{(1, 0, 0, 1)} \} \\ m, r, \theta \in \mathbb{R}$$

c) stud diag

$$v_{\lambda_1} = \text{sp} \{ v_1 \} \Rightarrow \dim v_{\lambda_1} = 1 = m a (\lambda_1)$$

$$X = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow \Delta_X = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = 1 \neq 0 \Rightarrow \text{rg } X = 3 \Rightarrow$$

$$v_{\lambda_2} = \text{sp} \{ v_2, v_3, v_4 \} \Rightarrow \dim v_{\lambda_2} = 3 = m a (\lambda_2)$$

$\Rightarrow A$ is diag

$$d) D = \begin{pmatrix} -2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

$$B = \{ v_1 = (-1, 1, 1, 1), v_2 = (1, 1, 0, 0), v_3 = (1, 0, 1, 0),$$

$$v_4 = (1, 0, 0, 1) \}$$

$$2. a) \quad A = \begin{pmatrix} 0 & 4 \\ 2 & 2 \end{pmatrix}$$

$$p(x) = \det(A - x I_2) = \begin{vmatrix} -x & 4 \\ 2 & 2-x \end{vmatrix} = x^2 - 2x - 8 = (x-4)(x+2)$$

$$p(x) = 0 \Rightarrow \begin{cases} x_1 = -2 & m_a(x_1) = 1 \\ x_2 = 4 & m_a(x_2) = 1 \end{cases} \quad \text{Spec } A = \{-2, 4\}$$

Subspații proprii:

$$S_{x_1=-2} \begin{cases} 2x + 4y = 0 \\ 2x + 4y = 0 \end{cases} \Rightarrow \begin{cases} 2x = -4y \\ x = -2y \end{cases} \quad \text{cu } (A - x I_2) = 0$$

$$\begin{cases} x \text{ nec } nr \\ y = x \text{ nec nec} \end{cases} \Rightarrow x = -2\alpha$$

$$\text{deci } v_{x_1} = \left\{ \alpha \begin{pmatrix} -2 \\ 1 \end{pmatrix} \mid \alpha \in \mathbb{R} \right\}$$

$$S_{x_2=4} \begin{cases} -4x + 4y = 0 \\ 2x - 2y = 0 \end{cases} \Rightarrow \begin{cases} 4x = 4y \\ x = y \end{cases}$$

$$\begin{cases} x \text{ nec } nr \\ y = \beta \text{ nec nec} \end{cases} \Rightarrow x = \beta$$

$$\text{deci } v_{x_2} = \left\{ \beta \begin{pmatrix} 1 \\ 1 \end{pmatrix} \mid \beta \in \mathbb{R} \right\}$$

$$b) \quad A = \begin{pmatrix} 3 & 4 \\ 0 & 5 \end{pmatrix}$$

$$p(x) = \det(A - x I_2) = \begin{vmatrix} 3-x & 4 \\ 0 & 5-x \end{vmatrix} = (x-5)(x-3)$$

$$p(x) = 0 \Rightarrow \begin{cases} x_1 = 3 & m_a(x_1) = 1 \\ x_2 = 5 & m_a(x_2) = 1 \end{cases}$$

$$S'_{\lambda_1=3} \begin{cases} 4y = 0 \\ 2y = 0 \end{cases} \Rightarrow \begin{matrix} y = 0 \\ \text{rang}(A - \lambda_1 I_3) = 1 \end{matrix}$$

$$\begin{cases} x = \alpha \text{ rec } \cancel{\text{rec}} \text{ rec} \Rightarrow \\ y = \text{rec } m \end{cases}$$

$$\text{decu: } V_{\lambda_1} = \{ \alpha (1, 0) \mid \alpha \in \mathbb{R} \}$$

$\uparrow$
 v_1

$$S'_{\lambda_2=5} \begin{cases} -2x + 4y = 0 \\ 0 + 0 = 0 \end{cases} \Rightarrow x = 2y$$

$$\begin{cases} x = \text{rec } m \\ y = \beta \text{ rec } m \end{cases} \Rightarrow x = 2\beta$$

$$\text{decu } \cancel{V_{\lambda_2}} = V_{\lambda_2} = \{ \beta (2, 1) \mid \beta \in \mathbb{R} \}$$

$\uparrow$
 v_2

$$1) \quad A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix}$$

$$p(x) = \det(A - xI_3) = \begin{vmatrix} 1-x & 0 & 0 \\ 0 & -x & 2 \\ 0 & 2 & -x \end{vmatrix} =$$

$$= -x^3 + x^2 + 4x - 4 = -x^2(x-1) + 4(x-1) = (-x^2+4)(x-1) =$$

$$= -(x^2-4)(x-1) = -(x+2)(x-2)(x-1)$$

$$p(x) = 0 \Leftrightarrow \begin{cases} \lambda_1 = -2, & m_a(\lambda_1) = 1 \\ \lambda_2 = 2, & m_a(\lambda_2) = 1 \\ \lambda_3 = 1, & m_a(\lambda_3) = 1 \end{cases}$$

Substitution procedure:

$$\begin{cases} \lambda_1 = -2 \\ \begin{cases} x = 0 \\ 2y + 2z = 0 \\ 2y + 2z = 0 \end{cases} \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = -z \end{cases}$$

$$A_{\lambda} = \begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix} = 2 \neq 0 \Rightarrow \text{rg}(A - \lambda_1 I_2) = 2$$

$$\begin{cases} x, y \text{ free var} \\ z = \alpha \text{ free var} \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = -\alpha \end{cases}$$

$$\text{dec } v_{\lambda_1} = \{ \alpha (0, -1, 1) \mid \alpha \in \mathbb{R} \}$$

$\uparrow$
 v_1

$$\begin{cases} \lambda_2 = 2 \\ \begin{cases} -x = 0 \\ -2y + 2z = 0 \\ 2y - 2z = 0 \end{cases} \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = z \end{cases}$$

$$A_{\lambda} = \begin{vmatrix} -1 & 0 \\ 0 & -2 \end{vmatrix} = 2 \neq 0 \Rightarrow \text{rg}(A - \lambda_2 I_2) = 2$$

$$\begin{cases} x, y \text{ free var} \\ z = \beta \text{ free var} \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = \beta \end{cases}$$

$$\text{dec } v_{\lambda_2} = \{ \beta (0, 1, 1) \mid \beta \in \mathbb{R} \}$$

$\uparrow$
 v_2

$$\begin{cases} \lambda_3 = 1 \\ \begin{cases} 0 = 0 \\ -y + 2z = 0 \\ 2y - z = 0 \end{cases} \end{cases} \Rightarrow \begin{cases} y = 2z \\ z = 2y \end{cases} \Rightarrow \begin{cases} y = 0 \\ z = 0 \end{cases}$$

$$A_{\lambda} = \begin{vmatrix} 1 & 2 \\ 2 & -1 \end{vmatrix} = -3 \neq 0 \Rightarrow \text{rg}(A - \lambda_3 I_2) = 2$$

$$\begin{cases} y, z \text{ free var} \\ x = 0 \text{ free var} \end{cases} \Rightarrow \begin{cases} y = 0 \\ z = 0 \end{cases}$$

$$\text{dec } v_{\lambda_3} = \{ \alpha (1, 0, 0) \mid \alpha \in \mathbb{R} \}$$

$\uparrow$
 v_3

$$d) A = \begin{pmatrix} 0 & 3 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$p(x) = \det(A - xI_3) = \begin{vmatrix} -x & 3 & 0 \\ 4 & -x & 0 \\ 0 & 0 & 3-x \end{vmatrix} = -x^3 + 3x^2 + 36x - 108$$

$$= -x^2(x-3) + 36(x-3) = -(x-36)(x-3) = -(x+6)(x-6)(x-3)$$

$$p(x) = 0 \Leftrightarrow \begin{cases} x_1 = -6 & m_a(x_1) = 1 \\ x_2 = 6 & m_a(x_2) = 1 \\ x_3 = 3 & m_a(x_3) = 1 \end{cases}$$

Subspații proprii

$$S_{x_1 = -6} \begin{cases} 6x + 3y = 0 \\ 4x + 6y = 0 \\ 3z = 0 \end{cases} \Rightarrow \begin{matrix} z = 0 \\ x = -\frac{3}{2}y \Rightarrow y = -\frac{2}{3}x \end{matrix}$$

$$A_{x_1} = \begin{vmatrix} 6 & 0 \\ 0 & 3 \end{vmatrix} = 18 \neq 0 \Rightarrow \dim(A - x_1 I_2) = 2$$

$$\begin{cases} y, z \text{ nec. } 0 \\ x = \alpha \end{cases} \Rightarrow \begin{matrix} z = 0 \\ y = -\frac{2}{3}\alpha \end{matrix}$$

$$\text{deci } v_{x_1} = \left\{ \alpha \left(1, -\frac{2}{3}, 0 \right) \mid \alpha \in \mathbb{R} \right\}$$

$$S_{x_2 = 6} \begin{cases} -6x + 3y = 0 \\ 4x - 6y = 0 \\ -3z = 0 \end{cases} \Rightarrow \begin{matrix} z = 0 \\ y = \frac{2}{3}x \end{matrix}$$

$$A_{x_2} = \begin{vmatrix} -6 & 0 \\ 0 & -3 \end{vmatrix} = 18 \neq 0 \Rightarrow \dim(A - x_2 I_2) = 2$$

$$\begin{cases} z, z \text{ rec pr} \\ x = \beta \end{cases} \Rightarrow \begin{cases} z = 0 \\ y = \frac{2}{-3} \beta \end{cases} \Rightarrow$$

$$\text{dec } v_{x_2} = \left\{ \beta \left(1, \frac{2}{3}, 0 \right) \mid \beta \in \mathbb{R} \right\}$$

$\uparrow$
 v_2

$$y_{x_3} = 3 \begin{cases} -3x + 3y = 0 \\ x - 3y = 0 \\ 0 = 0 \end{cases} \Rightarrow \begin{cases} x = 3y \\ y = \frac{1}{3}x \end{cases} \Rightarrow x = y = 0$$

$$\Delta x = \begin{vmatrix} -3 & 3 \\ 1 & -3 \end{vmatrix} = -2 \neq 0 \Rightarrow \text{rg}(A - x_3 I_3) = 2$$

$$\begin{cases} x, y \text{ rec pr} \\ z = \gamma \text{ rec rec} \end{cases} \Rightarrow \begin{cases} x = z = 0 \end{cases} \Rightarrow$$

$$\text{dec } v_{x_3} = \left\{ \gamma (0, 0, 1) \mid \gamma \in \mathbb{R} \right\}$$

$\uparrow$
 v_3

$$3. a) \quad v_{x_1} = \eta \langle v_1 \rangle \Rightarrow \dim v_{x_1} = 1 = m a(x_1)$$

$$v_{x_2} = \eta \langle v_2 \rangle \Rightarrow \dim v_{x_2} = 1 = m a(x_2)$$

$\Rightarrow$ A diag

$$D = \begin{pmatrix} -2 & 0 \\ 0 & 4 \end{pmatrix}$$

$$b) \quad v_{x_1} = \eta \langle v_1 \rangle \Rightarrow \dim v_{x_1} = 1 = m a(x_1)$$

$$v_{x_2} = \eta \langle v_2 \rangle \Rightarrow \dim v_{x_2} = 1 = m a(x_2)$$

$\Rightarrow$ A diag

$$D = \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix}$$

$$c) \quad V_{\lambda_1} = \ker \langle v_1 \rangle \Rightarrow \dim V_{\lambda_1} = 1 = m_a(\lambda_1)$$

$$V_{\lambda_2} = \ker \langle v_2 \rangle \Rightarrow \dim V_{\lambda_2} = 1 = m_a(\lambda_2)$$

$$V_{\lambda_3} = \ker \langle v_3 \rangle \Rightarrow \dim V_{\lambda_3} = 1 = m_a(\lambda_3)$$

$\Rightarrow$ A diag

$$D = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$d) \quad V_{\lambda_1} = \ker \langle v_1 \rangle \Rightarrow \dim V_{\lambda_1} = 1 = m_a(\lambda_1)$$

$$V_{\lambda_2} = \ker \langle v_2 \rangle \Rightarrow \dim V_{\lambda_2} = 1 = m_a(\lambda_2)$$

$$V_{\lambda_3} = \ker \langle v_3 \rangle \Rightarrow \dim V_{\lambda_3} = 1 = m_a(\lambda_3)$$

$\Rightarrow$ A diag

$$D = \begin{pmatrix} -6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

Subspațiile proprii:

$$S_{\lambda_1 = -1} \begin{cases} 2x + 4y = 0 \\ -3y + 3z = 0 \\ y + z = 0 \end{cases} \Rightarrow \begin{cases} 2x + 4y = 0 \\ y + z = 0 \end{cases} \Rightarrow \begin{cases} x = -2y \\ y = -z \end{cases}$$

$$\Delta x = \begin{vmatrix} 2 & 4 \\ 1 & 1 \end{vmatrix} = -2 \neq 0 \Rightarrow \text{rang}(A_T - \lambda_1 I_3) = 2$$

$$\begin{cases} x, y \text{ nec. pr.} \\ z = \alpha \text{ nec. rec.} \end{cases} \Rightarrow \begin{cases} x = -2\alpha \\ y = -\alpha \end{cases}$$

$$\text{deci } v_{\lambda_1} = \left\{ \alpha \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} \mid \alpha \in \mathbb{R} \right\}$$

$\overset{v_1}{\uparrow}$

$$S_{\lambda_2 = 1} \begin{cases} 4y = 0 \\ y + 3z = 0 \\ y - z = 0 \end{cases} \Rightarrow \begin{cases} y = 0 \\ z = 0 \end{cases}$$

$$\Delta x = \begin{vmatrix} 4 & 0 \\ 1 & -1 \end{vmatrix} = -4 \neq 0 \Rightarrow \text{rang}(A_T - \lambda_2 I_3) = 2$$

$$\begin{cases} y, z \text{ nec. pr.} \\ x = \beta \text{ nec. rec.} \end{cases} \Rightarrow$$

$$\text{deci } v_{\lambda_2} = \left\{ \beta \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \mid \beta \in \mathbb{R} \right\}$$

$\overset{v_2}{\uparrow}$

$$S_{\lambda_3 = 3} \begin{cases} -2x + 4y = 0 \\ -y + 3z = 0 \\ y - 3z = 0 \end{cases} \Rightarrow \begin{cases} x = 2y \\ y = 3z \end{cases}$$

$$\Delta x = \begin{vmatrix} -2 & 4 \\ -1 & 3 \end{vmatrix} = -2 \neq 0 \Rightarrow \text{rang}(A_T - \lambda_3 I_3) = 2$$

$$\begin{cases} x, y \text{ nec. pr.} \\ z = \gamma \text{ nec. rec.} \end{cases} \Rightarrow \begin{cases} x = 6\gamma \\ y = 3\gamma \end{cases}$$

$$\text{deci } v_{\lambda_3} = \left\{ \gamma \begin{pmatrix} 6 \\ 3 \\ 1 \end{pmatrix} \mid \gamma \in \mathbb{R} \right\}$$

$\overset{v_3}{\uparrow}$

$$4. \quad T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T(x, y, z) = (x + 4y, 2y + 3z, y)$$

$$a) \quad T(x) = Ax, \text{ unde } x = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \text{ in } A = \begin{pmatrix} 1 & 4 & 0 \\ 0 & 2 & 3 \\ 0 & 1 & 0 \end{pmatrix} \in M_3(\mathbb{R})$$

$$\text{fie } x_1, x_2 \in \mathbb{R}^3$$

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} \quad \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$$

$$\text{in } x_1, x_2 \in \mathbb{R}$$

$$\begin{aligned} T(x_1 x_2 + x_2 x_1) &= A(x_1 x_1 + x_2 x_2) = A(x_1 x_1) + A(x_2 x_2) = \\ &= (Ax_1) x_1 + (Ax_2) x_2 = (x_1 A) x_1 + (x_2 A) x_2 = x_1 (Ax_1) + x_2 (Ax_2) = \\ &= x_1 T(x_1) + x_2 T(x_2) \Rightarrow T \text{ apl lin} \end{aligned}$$

$$b) \quad A_T = \begin{pmatrix} 1 & 4 & 0 \\ 0 & 2 & 3 \\ 0 & 1 & 0 \end{pmatrix} \in M_3(\mathbb{R})$$

$$c) \quad p(\lambda) = \det(A_T - \lambda I_3) = \begin{vmatrix} 1-\lambda & 4 & 0 \\ 0 & 2-\lambda & 3 \\ 0 & 1 & -\lambda \end{vmatrix} =$$

$$= -\lambda^3 + 3\lambda^2 + \lambda - 3 = -\lambda^2(\lambda - 3) + \lambda - 3 = -(\lambda^2 - 1)(\lambda - 3)$$

$$= -(\lambda + 1)(\lambda - 1)(\lambda - 3)$$

$$p(\lambda) = 0 \Leftrightarrow$$

$$\begin{cases} \lambda_1 = -1 & m_a(\lambda_1) = 1 \\ \lambda_2 = 1 & m_a(\lambda_2) = 1 \\ \lambda_3 = 3 & m_a(\lambda_3) = 1 \end{cases}$$

$$S_{\text{spec}} = \{-1, 1, 3\}$$

$$d) \quad v_{\lambda_1} = \eta \langle v_1 \rangle \Rightarrow \dim v_{\lambda_1} = 1 = m a(\lambda_1)$$

$$v_{\lambda_2} = \eta \langle v_2 \rangle \Rightarrow \dim v_{\lambda_2} = 1 = m a(\lambda_2)$$

$$v_{\lambda_3} = \eta \langle v_3 \rangle \Rightarrow \dim v_{\lambda_3} = 1 = m a(\lambda_3)$$

$\Rightarrow A_T$ diag

$$e) \quad D = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$B = \{ v_1 = (2, -1, 1), v_2 = (1, 0, 0), v_3 = (6, 3, 1) \}$$

$$C = (v_1 \ v_2 \ v_3) = \begin{pmatrix} 2 & 1 & 6 \\ -1 & 0 & 3 \\ 1 & 0 & 1 \end{pmatrix}$$

GAZ - Temă seminar 9

1. $Q: \mathbb{R}^3 \rightarrow \mathbb{R}$

$$Q(x) = x_1^2 + 3x_2^2 + x_3^2 - 2x_1x_2 - 4x_2x_3 \quad (\forall) x = (x_1, x_2, x_3) \in \mathbb{R}^3$$

a) $A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 3 & -2 \\ 0 & -2 & 1 \end{pmatrix} \rightarrow$ matrice asociată pătră a în raport cu baza canonică

$$\begin{aligned} Q(x) &= (x_1^2 - 2x_1x_2 + x_2^2) + (4x_2^2 - 4x_2x_3 + x_3^2) - 2x_2^2 \\ &= (x_1 - x_2)^2 + (2x_2 - x_3)^2 - 2x_2^2 \end{aligned}$$

efectul schimbării de coordonate

$$\begin{cases} y_1 = x_1 - x_2 \\ y_2 = 2x_2 - x_3 \\ y_3 = x_2 \end{cases}$$

$\Rightarrow Q(x) = y_1^2 + y_2^2 - 2y_3^2, \quad (\forall) x = (y_1, y_2, y_3) \in \mathbb{R}^3$

$\downarrow$
coord în raport cu noua bază de raportare

b) $A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 3 & -2 \\ 0 & -2 & 1 \end{pmatrix} \rightarrow$ m. asociată lui Q_1

$\rightarrow \Delta_1 = 1$

$\Delta_2 = \begin{vmatrix} 1 & -1 \\ -1 & 3 \end{vmatrix} = 2$

$\Delta_3 = \det A = \begin{vmatrix} 1 & -1 & 0 \\ -1 & 3 & -2 \\ 0 & -2 & 1 \end{vmatrix} = -2$

$$\Rightarrow Q(x) = \frac{1}{\Delta_1} (x_1')^2 + \frac{\Delta_1}{\Delta_2} (x_2')^2 + \frac{\Delta_2}{\Delta_3} (x_3')^2$$

$$= \frac{1}{1} (x_1')^2 + \frac{1}{2} (x_2')^2 + 1 (x_3')^2, (\forall) x = (x_1', x_2', x_3') \in \mathbb{R}^3$$

$$2. \quad A = \begin{pmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 3 \\ 1 & 2 & -3 \end{pmatrix} \quad \Rightarrow \quad C^t = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 2 \\ 1 & 3 & -3 \end{pmatrix}$$

$$A' = C^T A C$$

$$= \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 2 \\ 1 & 3 & -3 \end{pmatrix} \begin{pmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 3 \\ 1 & 2 & -3 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -3 & -2 \\ 6 & -9 & 2 \\ -4 & 7 & -6 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 3 \\ 1 & 2 & -3 \end{pmatrix}$$

$$= \begin{pmatrix} -5 & -1 & -3 \\ -1 & 25 & -27 \\ -3 & -27 & 35 \end{pmatrix} \quad \rightarrow \text{in axes of Eulerian system in rapport in new basis}$$

3. fie $F: \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}$

$$F(x, y) = 2x_1y_1 + x_2y_2 - 2x_1y_2 - 2x_2y_1 - 2x_2y_3 - 2x_3y_2$$

a) $x = (x_1, x_2, x_3) \in \mathbb{R}^3$

$y = (y_1, y_2, y_3) \in \mathbb{R}^3$

a, b) dem linear lui F in ambele arg:

fie $x, y, z \in \mathbb{R}^3$, $x = (x_1, x_2, x_3)$

in $\alpha, \beta \in \mathbb{R}$ $y = (y_1, y_2, y_3)$

$z = (z_1, z_2, z_3)$

$$\begin{aligned} \text{I} \quad F(\alpha x + \beta y, z) &= 2(\alpha x_1 + \beta y_1)z_1 + (\alpha x_2 + \beta y_2)z_2 - 2(\alpha x_1 + \beta y_1)z_2 - \\ &- 2(\alpha x_2 + \beta y_2)z_1 - 2(\alpha x_2 + \beta y_2)z_3 - 2(\alpha x_3 + \beta y_3)z_2 = \\ &= \alpha(2x_1z_1 + x_2z_2 - 2x_1z_2 - 2x_2z_1 - 2x_2z_3 - 2x_3z_2) + \\ &+ \beta(2y_1z_1 + y_2z_2 - 2y_1z_2 - 2y_2z_1 - 2y_2z_3 - 2y_3z_2) \\ &= \alpha F(x, z) + \beta F(y, z) \end{aligned}$$

$$\begin{aligned} \text{II} \quad F(x, \alpha y + \beta z) &= 2x_1(\alpha y_1 + \beta z_1) + x_2(\alpha y_2 + \beta z_2) - 2x_1(\alpha y_2 + \beta z_2) - \\ &- 2x_2(\alpha y_1 + \beta z_1) - 2x_2(\alpha y_3 + \beta z_3) - 2x_3(\alpha y_2 + \beta z_2) = \\ &= \alpha(2x_1y_1 + x_2y_2 - 2x_1y_2 - 2x_2y_1 - 2x_2y_3 - 2x_3y_2) + \\ &+ \beta(2x_1z_1 + x_2z_2 - 2x_1z_2 - 2x_2z_1 - 2x_2z_3 - 2x_3z_2) \\ &= \alpha F(x, y) + \beta F(x, z) \end{aligned}$$

$\Rightarrow$ F liniară în ambele arg

$A = \begin{pmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{pmatrix}$ - matricea lui F în raport cu baza canonică

$A \rightarrow$ matrice sim (${}^t A = A$) $\Rightarrow F$ formă bil sim

$$c) B_1 = \{(1, 1, 1), (2, -1, 2), (1, 3, -3)\}$$

$$C = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 3 \\ 1 & 2 & -3 \end{pmatrix} \Rightarrow {}^t C = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 2 \\ 1 & 3 & -3 \end{pmatrix}$$

matrice bil sim in rep cu baza B_1 : $A' = {}^t C A C$

$$A' = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 2 \\ 1 & 3 & -3 \end{pmatrix} \begin{pmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 3 \\ 1 & 2 & -3 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -3 & -2 \\ 6 & -9 & 2 \\ -4 & 7 & -6 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 3 \\ 1 & 2 & -3 \end{pmatrix}$$

$$= \begin{pmatrix} -5 & -1 & -3 \\ -1 & 25 & -27 \\ 3 & -27 & 35 \end{pmatrix} \rightarrow \text{m arec' bil sim in rep cu noua bază}$$

d) $Q: \mathbb{R}^3 \rightarrow \mathbb{R}$ prin met dedublării :

$$Q(x) = 2x_1^2 + x_2^2 - 4x_1x_2 - 4x_2x_3 \quad (x) = (x_1, x_2, x_3) \in \mathbb{R}^3$$

i formă canonică pătratică - Gauss :

$$\begin{aligned} Q(x) &= \cancel{(4x_1^2 - 4x_1x_2 + x_2^2)} + (4x_2^2 - 4x_2x_3 + x_3^2) - \cancel{2x_1^2 - 4x_1^2 - x_3^2} \\ &= (2x_1 - x_2)^2 + (2x_2 - x_3)^2 - 2x_1^2 - 4x_1^2 - x_3^2 \\ &= (2x_1^2 - 4x_1x_2) + x_2^2 - 4x_2x_3 \\ &= (\sqrt{2}x_1 - \sqrt{2}x_2)^2 - x_2^2 - 4x_2x_3 \end{aligned}$$

$$Q(x) = (\sqrt{2}x_1 - \sqrt{2}x_2)^2 - (x_2^2 + 4x_2x_3) \\ = (\sqrt{2}x_1 - \sqrt{2}x_2)^2 - (x_2 + 2x_3)^2 + 4x_3^2$$

această schimb de coordonate

$$\begin{cases} y_1 = \sqrt{2}x_1 - \sqrt{2}x_2 \\ y_2 = x_2 + 2x_3 \\ y_3 = x_3^2 \end{cases}$$

$$\Rightarrow Q(x) = y_1^2 - y_2^2 + 4y_3^2 \quad (\forall x = (y_1, y_2, y_3) \in \mathbb{R}^3)$$

↓
word în raport cu noua bază de raportare

forma canonică pătratică - Jordan

$$\begin{cases} \lambda_1 = 2 \\ \lambda_2 = \begin{vmatrix} 2 & -2 \\ -2 & 1 \end{vmatrix} = -2 \\ \lambda_3 = -8 \end{cases} \quad \left| \lambda_i \neq 0, (\forall) i = \overline{1, 3} \right.$$

$$\Rightarrow Q(x) = \frac{1}{2}(x_1')^2 - (x_2')^2 + \frac{1}{4}(x_3')^2, \quad (\forall x = (x_1', x_2', x_3'))$$

↓
word în raport cu noua bază de raportare

$$1. \quad B = \{ v_1 = (1, 1, 1), v_2 = (1, 1, -1), v_3 = (1, -1, -1) \}$$

$$e_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{3}} (1, 1, 1)$$

$$e_2' = v_2 - \langle v_2, e_1 \rangle e_1$$

$$= (1, 1, -1) - \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}} (1, 1, 1)$$

$$= (1, 1, -1) - \frac{1}{3} (1, 1, 1) = \frac{2}{3} (1, 1, -2) \Rightarrow \|e_2'\| = \frac{2}{3} \sqrt{6}$$

$$e_2 = \frac{e_2'}{\|e_2'\|} = \frac{1}{\sqrt{6}} (1, 1, -2)$$

$$e_3' = v_3 - \langle v_3, e_1 \rangle e_1 - \langle v_3, e_2 \rangle e_2$$

$$= (1, -1, -1) + \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}} (1, 1, 1) - 2 \cdot \frac{1}{\sqrt{6}} \cdot \frac{1}{\sqrt{6}} (1, 1, -2)$$

$$= (1, -1, -1) + \frac{1}{3} (1, 1, 1) - \frac{2}{3} (1, 1, -2) = (1, -1, 0) \Rightarrow \|e_3'\| = \sqrt{2}$$

$$e_3 = \frac{e_3'}{\|e_3'\|} = \frac{1}{\sqrt{2}} (1, -1, 0)$$

$$\text{Adăugăm } \{ e_1, e_2, e_3 \} \in \mathbb{R}^3$$

baza ortonormată obținută prin p.o.g.s. din baza dată

$$2. \quad B' = \{ e_1 = \frac{1}{\sqrt{2}} (0, 1, 1), e_2 = \frac{1}{\sqrt{6}} (2, -1, 1), e_3 = \frac{1}{\sqrt{3}} (1, 1, -1) \}$$

$$w = (-1, 1, 2)$$

scriem vectorul w în baza B' dată:

$$w = w_1 e_1 + w_2 e_2 + w_3 e_3$$

$$\langle w, e_1 \rangle = w_1 \underbrace{\langle e_1, e_1 \rangle}_{=1} + w_2 \underbrace{\langle e_2, e_1 \rangle}_{=0} + w_3 \underbrace{\langle e_3, e_1 \rangle}_{=0}$$

$$\Rightarrow w_1 = \langle w, e_1 \rangle = \frac{3}{\sqrt{2}}$$

$$\langle w, e_2 \rangle = w_1 \underbrace{\langle e_1, e_2 \rangle}_{=0} + w_2 \underbrace{\langle e_2, e_2 \rangle}_{=1} + w_3 \underbrace{\langle e_3, e_2 \rangle}_{=0}$$

$$\Rightarrow w_2 = \langle w, e_2 \rangle = \frac{(-2 - 1 + 2)}{\sqrt{6}} = -\frac{1}{\sqrt{6}}$$

$$\langle w, e_3 \rangle = w_1 \underbrace{\langle e_1, e_3 \rangle}_{=0} + w_2 \underbrace{\langle e_2, e_3 \rangle}_{=0} + w_3 \underbrace{\langle e_3, e_3 \rangle}_{=1}$$

$$\Rightarrow w_3 = \langle w, e_3 \rangle = \frac{(-1 + 1 - 2)}{\sqrt{3}} = -\frac{2}{\sqrt{3}}$$

$$3. \quad B = \{ v_1 = (2, 2, 1), v_2 = (-2, -1, 2), v_3 = (5, -6, 2) \}$$

$$e_1 = \frac{v_1}{\|v_1\|} = \frac{1}{3} (2, 2, 1)$$

$$e'_2 = v_2 - \langle v_2, e_1 \rangle e_1$$

$$= (-2, -1, 2) - \frac{(-4)}{3} \cdot \frac{1}{3} (2, 2, 1)$$

$$= \left(-\frac{10}{9}, -\frac{1}{9}, \frac{22}{9} \right) = \frac{1}{9} (-10, -1, 22)$$

$$\|e'_2\| = \frac{1}{9} \cdot \sqrt{100 + 1 + 484} = \frac{1}{9} \cdot \sqrt{585} = \frac{1}{3} \cdot \sqrt{65}$$

$$e_2 = \frac{e'_2}{\|e'_2\|} = \frac{1}{\sqrt{65}} (-10, -1, 22)$$

$$e'_3 = v_3 - \langle v_3, e_1 \rangle e_1 - \langle v_3, e_2 \rangle e_2$$

$$= (5, -6, 2) - 0 \cdot e_1 - 0 \cdot e_2$$

$$= (5, -6, 2)$$

$$\|e'_3\| = \sqrt{25 + 36 + 4} = \sqrt{65}$$

$$e_3 = \frac{e'_3}{\|e'_3\|} = \frac{1}{\sqrt{65}} (5, -6, 2)$$

Având $\{e_1, e_2, e_3\} \subset \mathbb{R}^3$

bază ortonormală sît, prin p.p.s. din baza dată.

$$4. a) U = \langle \underbrace{(1, 2, 1)}_{u_1}, \underbrace{(1, -1, 2)}_{u_2} \rangle$$

$$U^\perp = \{ y \in \mathbb{R}^3 / y \perp x, (\forall) x \in U \}$$

$$B = \{ u_1, u_2 \} \subset U$$

bază

$$\text{Scrie } y = (y_1, y_2, y_3) \in \mathbb{R}^3 \text{ a } \hat{=} \begin{cases} y \perp u_1 \\ y \perp u_2 \end{cases} \Leftrightarrow \begin{cases} \langle y, u_1 \rangle = 0 \\ \langle y, u_2 \rangle = 0 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} y_1 + 2y_2 + y_3 = 0 \\ y_1 - y_2 + 2y_3 = 0 \end{cases}$$

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 2 \end{pmatrix}$$

$$\Delta = \begin{vmatrix} 1 & 2 \\ 1 & -1 \end{vmatrix} = -3 \neq 0 \Rightarrow \text{rang } A = 2$$

$$\begin{cases} y_1, y_2 \text{ nec nec} \\ y_3 = \alpha \text{ nec nec} \end{cases} \Rightarrow \begin{cases} y_1 + 2y_2 = -\alpha \\ y_1 - y_2 = -2\alpha \end{cases} \Rightarrow \begin{cases} y_2 = \frac{\alpha}{3} \\ y_1 = -\frac{5\alpha}{3} \end{cases}$$

$$\text{deci } y = \alpha \left(-\frac{5}{3}, \frac{1}{3}, 1 \right), \alpha \in \mathbb{R}$$

$$U^\perp = \left\{ \alpha \left(-\frac{5}{3}, \frac{1}{3}, 1 \right) / \alpha \in \mathbb{R} \right\} = \left\langle \left(-\frac{5}{3}, \frac{1}{3}, 1 \right) \right\rangle$$

$\Rightarrow$ complementul ortogonal al lui U (sp. vect 1-dim)

$$b) V = \langle \underbrace{(2, -3, 1)}_v \rangle$$

$$V^\perp = \{ y \in \mathbb{R}^3 / y \perp x, (\forall) x \in V \}$$

$$B = \{ v \} \subset V$$

bază

fie $y = (y_1, y_2, y_3) \in \mathbb{R}^3$ a $\hat{c} \quad y \perp v \Leftrightarrow \langle y, v \rangle = 0 \Leftrightarrow$

$$\Leftrightarrow 2y_1 - 3y_2 + y_3 = 0$$

$$\text{deci } v^\perp = \{ y \in \mathbb{R}^3 \mid 2y_1 - 3y_2 + y_3 = 0 \}$$

(y_1, y_2, y_3)

→ mul orthogonal al lui v (sp. vect 2-dimensional)